

# Node.js: Building Your First CLI Tool

Your Command Center for School Tasks | Comprehensive Lesson Guide

## LESSON 1 What is a Command Line Interface (CLI)?

A **Command Line Interface (CLI)** allows you to interact directly with your computer system by typing text commands instead of clicking icons with a mouse (Graphical User Interface or GUI).



### GUI (Graphical User Interface)

Like a TV remote control with set buttons.  
Visual, user-friendly, but limited to provided options.



### CLI (Command Line Interface)

Like typing secret codes or magic spells!  
Fast, powerful, precise, and gives direct control over computer tasks.

## LESSON 2 Meet NPM – Your Magic Package Store

**NPM (Node Package Manager)** is like a superstore full of pre-written tools and libraries that developers use to extend their Node.js applications effortlessly.

### Popular Packages Include:

- **chalk** : Adds vibrant colors and formatting to terminal outputs.
- **inquirer** : Creates interactive CLI prompts and questions.
- **express** : Powers web server backends.
- **axios** : Fetches live data from internet APIs (like weather or scores).
- **figlet** : Turns plain text into large ASCII art banners.

## LESSON 3 Project Setup with `npm init`

To start any Node.js project, you first initialize it to track your packages and project configuration.

1. Open your Terminal or Command Prompt.
2. Navigate to your project directory using the change directory command: `cd my-cli-project`
3. Run the initialization command:

```
npm init -y
```

This generates a `package.json` file — essentially your project's official ID card!

## LESSON 4 Handling Arguments with `process.argv`

How does a command line script read what you type into the terminal? Node.js uses `process.argv`, acting like a postman delivering user input directly into your code logic.

Create a script named `assistant.js`:

```
// assistant.js
const name = process.argv[2];
console.log(`Hello, ${name}! Welcome to your Command Center.`);
```

Running the script:

```
node assistant.js Priya
```

Output: **Hello, Priya! Welcome to your Command Center.**

## LESSON 5 Adding Neon Colors with `chalk`

Make your terminal scripts engaging and colorful! The `chalk` package allows you to style your outputs with neon colors, bold text, and backgrounds.

1. Install the chalk package:

```
npm install chalk
```

2. Use chalk in your code:

```
const chalk = require('chalk');

console.log(chalk.green('Homework Task Added Successfully!'));
console.log(chalk.bold.yellow('Priority: High'));
```

## LESSON 6 Fun Ideas to Build Your Own CLI

Here are several creative, practical CLI project ideas you can build for daily tasks:

- **Pincode Finder:** Look up city details by postal code.
- **Cricket Score Notifier:** Live updates right in your console.
- **Vedic Maths Quiz:** Rapid mental math challenges.
- **Sanskrit Word Helper:** Discover daily vocabulary.
- **Festival Countdown:** Track days until Diwali or Holi.
- **Tiffin Menu Planner:** Randomize and organize lunch plans.

# Student Practice Worksheet

Test Your Knowledge: Node.js CLI & Fundamentals

**Instructions:** Answer the following practice questions based on the lesson materials provided. Write your answers clearly on a separate sheet of paper or in your notebook.

## Question 1: CLI vs GUI Concepts

Explain in your own words the primary difference between a Graphical User Interface (GUI) and a Command Line Interface (CLI). Why do software developers often prefer CLI tools for repetitive tasks?

## Question 2: Project Initialization

Which command is used to initialize a new Node.js project automatically without answering all setup prompts? What file does this command create, and what is its purpose?

## Question 3: Understanding Command Line Arguments

Consider the following command typed into the terminal:

```
node helper.js Math Homework 5
```

In Node.js, what values will be contained in `process.argv[0]`, `process.argv[1]`, and `process.argv[2]`?

## Question 4: Package Management & Styling

Name the NPM package discussed in the lesson used for adding custom colors to terminal outputs. Write a short 2-line Node.js snippet demonstrating how to import this package and print the word "SUCCESS" in bright green text.

## Question 5: Application Design

Choose one of the project ideas mentioned (e.g., Festival Countdown or Tiffin Menu Planner) or propose your own CLI tool idea. Describe:

- What command arguments the user would type in.
- How the script would process those arguments.

- What output would be displayed to the terminal.